



Tutorial 5 Mini-Lecture: Terrestrial Planets

The concepts behind Worksheet 5 • Lectures 9–10

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EnVision



Three Venus-bound missions over a Magellan radar mosaic. Credit: Widemann et al. (2023), SSRv 219, 56

What Worksheet 5 trains

Five problems, all built on Lectures 9–10. The physics you need, per problem:

1. **Equilibrium temperature and the greenhouse:** radiative balance, the one-layer greenhouse factor, the gap to the true surface temperature.
2. **The runaway greenhouse:** the vapour-pressure photosphere, the Simpson-Nakajima ceiling on outgoing infrared, which planet crosses it.
3. **Deuterium and Venus's lost water:** Rayleigh distillation, the D/H clock, the size of the vanished ocean.
4. **Thermal escape from Mars:** the escape speed, the Jeans parameter, why light molecules leave first.
5. **Mercury and Mars: two ways to end:** the spin-orbit resonance, thermal contraction, Phobos below the synchronous orbit.

Every problem has the shape and level of one exam question. All parts are analytical; no computer is needed.

Equilibrium temperature and the greenhouse

Radiative balance from Lecture 9:

$$T_{\text{eq}} = \left[\frac{S(1 - A)}{4\sigma} \right]^{1/4} \quad T_s = 2^{1/4} T_{\text{eq}}$$

- The factor $1/4$ is geometry: a disc of area πR^2 intercepts, a sphere of area $4\pi R^2$ radiates.
- One opaque layer warms the surface by exactly $2^{1/4} \approx 1.19$; the true greenhouse is larger.

Same law, two planets

Venus intercepts 1.91 times Earth's flux yet reflects most of it. Run both planets through the one formula, then measure each equilibrium temperature against the observed surface. That gap is the greenhouse.

One radiative balance sets the baseline temperature; the distance from it to the surface is the greenhouse you must explain.

The runaway greenhouse

The infrared ceiling of Lecture 9:

$$F_{\text{OLR}}^{\text{max}} = \sigma T_{\text{phot}}^4 \quad p_{\text{phot}} = \frac{g}{\kappa}$$

- Clausius-Clapeyron fixes T_{phot} at the vapour-pressure photosphere, independent of the surface below.
- A hotter surface cannot raise the outgoing flux: it caps at the Simpson-Nakajima limit.

A saturated steam atmosphere has a fixed ceiling on its outgoing infrared; exceed it with sunlight, and no ocean survives.

Cross the ceiling, run away

If the absorbed sunlight $F_{\text{abs}} = S(1 - A)/4$ exceeds this cap, no steady surface temperature can balance it and the ocean evaporates. Test F_{abs} against the cap for both planets.

Deuterium and Venus's lost water

Rayleigh distillation of Lecture 9:

$$\boxed{\frac{R}{R_0} = f^{\alpha-1}} \quad f = \left(\frac{R}{R_0} \right)^{1/(\alpha-1)}$$

- f is the surviving hydrogen fraction, $R/R_0 = 150$ the measured deuterium enrichment, $\alpha < 1$ the fractionation factor.
- Invert for f , then divide today's water by f to recover the initial inventory.

How much can it prove?

Diffusion-limited escape ($\alpha = 0.5$) fractionates most efficiently and gives the smallest initial water, about 0.17 Earth ocean. A weaker model needs far more water for the same enrichment. The isotope ratio sets a lower bound only.

The deuterium enrichment is a lower bound on Venus's lost water, of order a tenth of an Earth ocean, not proof of a full ocean.

Thermal escape from Mars

The escape parameter of Lecture 10:

$$\lambda = \frac{GMm}{k_B T_{\text{exo}} r_{\text{exo}}} \quad v_{\text{esc}} = \sqrt{\frac{2GM}{r_{\text{exo}}}}$$

- λ is the ratio of gravitational to thermal energy at the exobase; it scales linearly with molecular mass.
- The Jeans flux carries $(1 + \lambda) e^{-\lambda}$ and a thermal-speed factor $\propto m^{-1/2}$; the exponential decides the outcome.

Light first, heavy never

Atomic hydrogen ($\lambda \approx 5.3$) escapes about 150 times faster than H_2 ($\lambda \approx 10.6$). Carbon dioxide ($\lambda \approx 233$) is suppressed by e^{-233} : thermal escape cannot remove it at all.

One exponential in the mass sorts which gases a planet keeps; Mars loses its hydrogen and holds its CO_2 .

Mercury and Mars: two ways to end

Three independent clocks:

$$\boxed{\frac{1}{T_{\text{solar}}} = \frac{1}{P_{\text{rot}}} - \frac{1}{P_{\text{orb}}}}$$
$$\frac{\Delta A}{A} = 2 \frac{\Delta R}{R} \quad T = 2\pi \sqrt{\frac{a^3}{GM}}$$

- Mercury's 3 : 2 resonance makes a solar day last two Mercury years.
- Cooling shrank Mercury by $\Delta R \approx 7$ km, a 0.57% area loss taken up as lobate scarps.

Above or below synchronous?

Phobos orbits in 7.7 h, faster than Mars turns (24.62 h), so it is below the areostationary radius. The tidal bulge it raises lags and drags it inward, so Phobos spirals down.

Size sets the cooling rate, cooling sets the dynamo lifetime, and gravity sets atmospheric retention; the same three controls produced Mercury and Mars.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** (“show that ...”) to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Most parts close by asking what the number means: that interpretation is graded, not only the arithmetic.

Worksheet and full solutions: ips.formingworlds.space/worksheets.html.
We discuss questions, not presentations of the solutions.